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Versuch: Properties of Cosmic Myons

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Exercises

1. Determination of the noise threshold of the detectors
2. Recording a pulse height spectrum
 - (a) Determination of suitable thresholds for muons and electrons/positrons
 - (b) Calibrating the pulse heights of the individual modules
3. Determination of the efficiency of the individual detector modules
4. Determination of the lifetime of the muon
5. Proof of the precession of the muon spin in the magnetic field and determination of the statistical significance

1 Theory

1.1 Composition of the cosmic rays

Primary cosmic rays consist mainly of protons (85 %), α -particles (12 %) and small portions of heavier nuclei, electrons and γ -quanta. Their energy ranges from 10^6 eV to 10^{21} eV. The spectrum can be described by the formula:

$$N(> E) = K \cdot \left(\frac{E}{E_0} \right)^{-\gamma}$$

with $E_0 = 10^{16}$ eV, $K = 3 \cdot 10^{-4} \text{ m}^{-2} \text{ h}^{-1}$ and $\gamma = 1.7 \dots 2.1$. $N(> E)$ is the rate of particles with energy greater than E , and γ is the spectral index.

When scientists observe cosmic radiation, they usually do not measure the primary particles directly, as these collide with atmospheric nuclei and thereby create secondary particles, which can then be detected.

Reactions in the atmosphere Primary nucleons interact via the strong interaction with air nuclei and create secondary particles such as nucleons and pions. The nucleons undergo further collisions, transferring part of the kinetic energy from the primary particles. This shower of particles, the so-called nuclear cascade, stops when the energy of the hadrons becomes too low to produce pions (~ 300 MeV). The neutral pions are unstable and quickly decay into two γ -quanta:

$$\pi^0 \rightarrow 2\gamma \quad \text{with} \quad \tau = 8.4 \cdot 10^{-17} \text{ s}$$

These γ -quanta initiate the electromagnetic cascade. They are so energetic that pair production (electron-positron pairs) becomes the most probable reaction. Both leptons acquire nearly the entire γ -energy and travel in approximately the same direction as the γ -quantum, with some angular deviation.

There are two main mechanisms by which leptons lose energy: ionization ($\propto Z \cdot \ln(E)$) and bremsstrahlung ($\propto Z^2 \cdot E$). At high energies, bremsstrahlung dominates, forming a cone-shaped shower of leptons and γ -quanta around the path of the primary particle. An energy greater than 10^{12} eV is required for such electromagnetic showers to reach the Earth's surface.

Decay of charged pions Charged pions decay via the weak interaction into muons and muon neutrinos ($\tau = 2.60 \cdot 10^{-8}$ s):

$$\pi^+ \rightarrow \mu^+ + \nu_\mu$$

$$\pi^- \rightarrow \mu^- + \bar{\nu}_\mu$$

It is important to note that the average lifetime of a charged pion is significantly longer than that of a neutral pion. However, hardly any charged pions reach the Earth's surface, as their mean free path is not sufficient. Muons, on the other hand, have a small cross-section and interact primarily via the electromagnetic interaction. Therefore, they can reach the Earth's surface and even penetrate deep below it. But muons can also further decay into an electron, electron-neutrino and muon-neutrino ($\tau = 2.19 \cdot 10^{-6}$ s):

$$\mu^+ \rightarrow e^+ + \nu_e + \bar{\nu}_\mu$$

$$\mu^- \rightarrow e^- + \bar{\nu}_e + \nu_\mu$$

One notices that the lifetime of a muon is even longer than that of a charged pion. The radiation observed at the Earth's surface is split into two components:

- muons $\rightarrow$ long-range, hard component
- electrons, positrons, bremsstrahlung quanta $\rightarrow$ short-range, soft component

A paradox arises due to the fact that the classical decay length of muons is only $\lambda = 600$ m. As muons are clearly observed at the Earth's surface, this seems contradictory. However, special relativity resolves this issue:

$$\text{mean lifetime on Earth: } \tau_e = \frac{\tau}{\sqrt{1 - \frac{v^2}{c^2}}} = \gamma \tau$$

Here, γ is the Lorentz factor. For an extremely relativistic particle, e.g., $E = 10$ GeV, this leads to $\lambda = 60$ km.

1.2 The deceleration of muons in matter

In matter muons lose energy due to bremsstrahlung and if the energy is smaller than GeV range ionization becomes relevant. At near-rest-energies, negative and positive muons behave differently.

Negative muons Negative muons are captured by the atoms in excited states and they land relatively quickly in the K-shell of the atom under emission of a X-rays. After that the muon is captured by the nucleus which opens up a second decay channel for μ^- . Hence the average lifetime is lowered and therefore one mostly detects μ^+ .

Positive muons At low energies μ^+ capture electrons from shells and form muonium (hydrogen structure). The repetition of the process of ionization and rebuilding muonium and also collisions until thermalization decelerates the muon.

The time scale of these processes are short compared to the lifetime of a muon.

- deceleration due to scattering: 10^{-10} s
- capture and ionization of muonium: $5 \cdot 10^{-13}$ s
- Thermalization: 10^{-12} s

1.3 Muon polarization

The charged Pion decay via the weak interaction into $\mu^{+/-}$ s and a $\nu_{\mu^{-/+}}$. Since Neutrinos have a neglectable mass their helicity and chirality is the same. In the standard model Neutrinos only have a negative chirality (anti Neutrinos have positive chirality). This implies that the helicity of the $\mu^{+/-}$ must be the same as helicity of the $\mu^{+/-}$ neutrino emitted in the same interaction. The only possibility for an unpolarized Muon stream is if it is equally likely to detect Muons that were emitted in the same direction as the momentum of the pion as it is to detect Muons that were emitted in the opposite direction. But this is not the case, since the Pions are not at rest compared to the detector on earth. For the Muons which were emitted in the opposite direction compared to the momentum of the Pion in the rest frame of the detector a higher energy is required for them to still be able to reach the detector. Since the spectrum of the pions is not constant, the Muons arriving at the detector will (on average) be polarized.

1.4 A proof of muon decay

The decay per energy and solid angle of the μ^+ into a Positronium and a anti electron neutrino has the following form:

$$\frac{d^2N}{d\epsilon d\Omega} = a(1 + b \cos \theta) \quad (1.1)$$

where θ is the angle between the spin and the momentum of the μ^+ . Hence the emitted positrons have a preferred direction which is along side the spin direction since cosine is maximal at an angle of 0.

1.5 The precession of muons in the magnetic field

The magnetic moment associated with the Spin of the Muons is $\vec{\mu} = \gamma \vec{J}$. In the presence of a magnetic field this leads to a precession along the magnetic field lines with an angular frequency of $\omega = \gamma B$ where $\gamma = \frac{ge}{2m_e}$ is the gyromagnetic ratio of the particle. The Lande factor g is predicted to be 1 by classical considerations. Dirac theory predicts that it is two in first order which is also the measured value.

2 Experimental setup

The experimental setup consists of four scintillators one copper plate and a solenoid arranged as shown in Figure 2.1

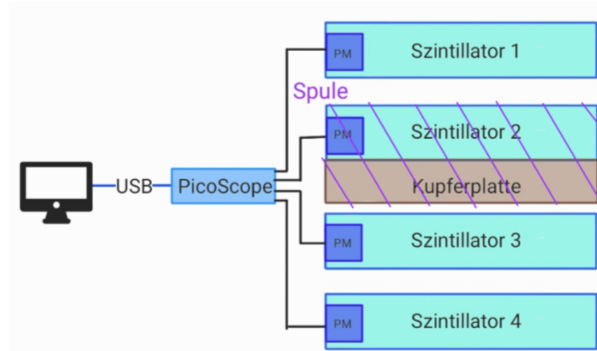


Figure 2.1: experimental setup

The signals from the scintillators are read out with a PicoScope. The copper plate "filters" the Muons, the anti Muons are much less affected by the copper. The solenoid around the second scintillator makes the spins precess which affects the decay of the muons.

3 Results

3.1 Panel efficiency

The efficiency of the panels can be calculated in the following way: All events in which e.g. scintillator 1 and 3 were triggered would suggest that scintillator 2 should have been triggered as well. This works fine for scintillators 2 and 3. For scintillators 1 and 4 the method has to be adjusted a bit. For e.g. scintillator 1 all events are taken into account in which at least two of the other scintillators were triggered. The problem is that this could have been triggered by a muon which is approaching at a slight angle in such a way that it never passed through scintillator 1. For scintillator 4 the decay of the muon is another effect which slightly changes the efficiency. These effects are not taken into account in this experiment.

First the data is inspected to make sure all panels and the trigger work properly.

As seen in Figure 3.1 the event rate is approximately constant which is good.

The pulse height spectrum of every individual panels is shown in Figure 3.2. Since they all have approximately the same peak height they all seem to work fine. Note that the peaks at zero for Panels B - D are caused by muons which were detected by panel A (and therefore an event was triggered) but decayed since then or passed at such an angle through the detector that the other panels were never triggered.

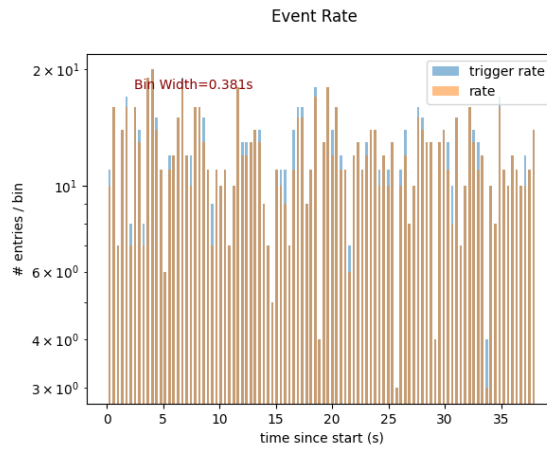


Figure 3.1: rate of events in the measurement used to calculate the efficiency of the panels

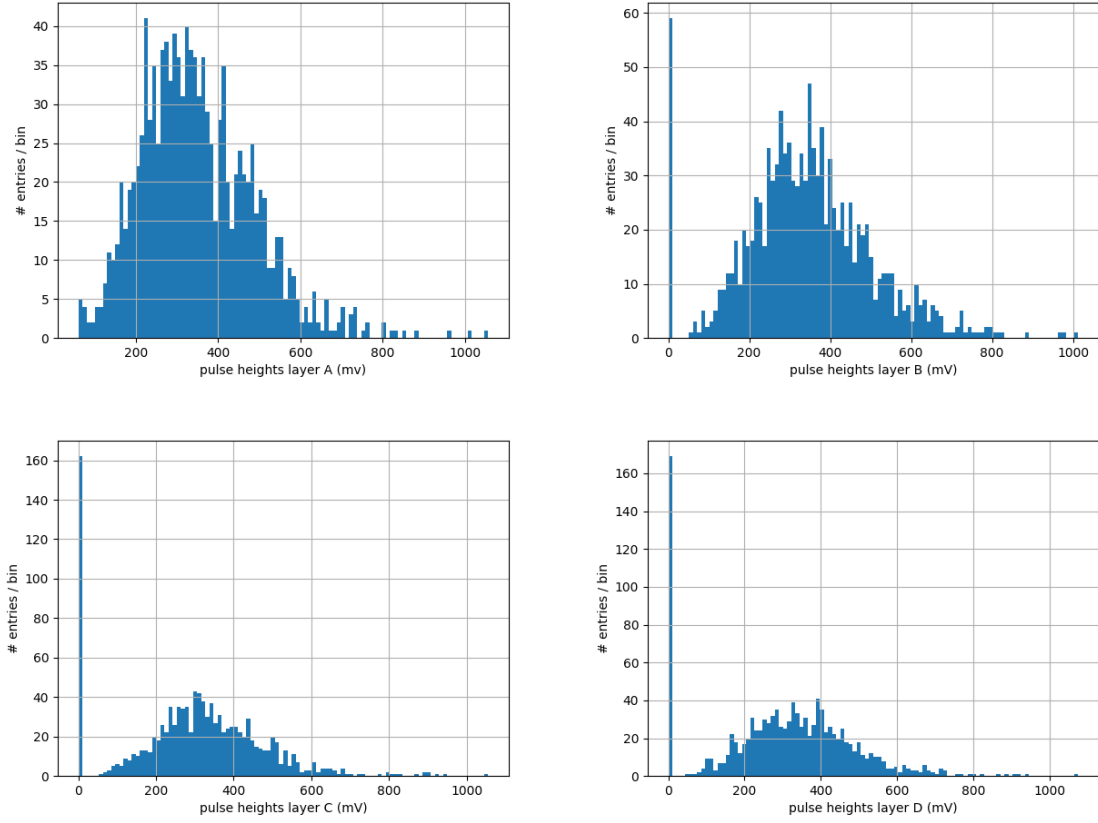


Figure 3.2: Pulse height spectrum of scintillator 1, 2, 3, 4 (top left to bottom right; note the different y-axis scales)

For the scintillators the following efficiencies were calculated

scintillator	efficiency
1	0.994
2	0.941
3	0.952
4	0.945

Table 3.1: Panel efficiency

3.2 Lifetime

The lifetime measurement works the following way: As soon as a muon was detected in scintillator 1 and 2 but not in scintillator 3 and 4 it can be assumed that the muon decayed inside the copper plate. This event starts the beginning of a lifetime measurement. An event where scintillator 2 or 2 and 1 but not 3 or 3 and 4 (for a positron which is emitted upwards) or 3 or 3 and 4 but not 2 or 2 and 1 (for a positron which is emitted downwards) are triggered marks the end of a lifetime measurement. A lifetime measurement signal is shown in Figure 3.3. The second peak is a signal from scintillator 3 and 4 which means that the positron was emitted downwards.

For the measurement of the lifetime about 250,000 events were recorded out of which 331 satisfied the criteria to resemble a decay with which the lifetime could be determined. The 331 lifetimes

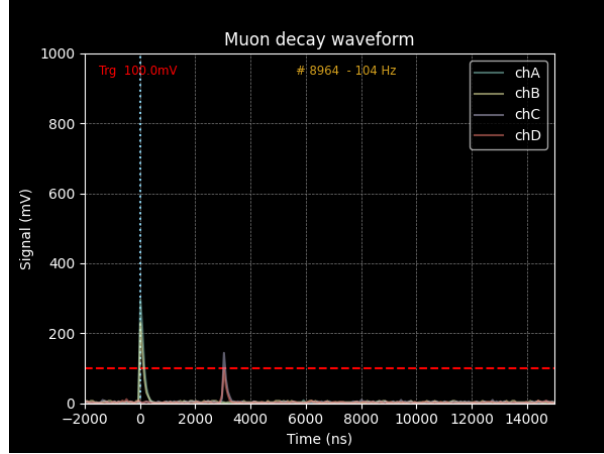


Figure 3.3: Caption

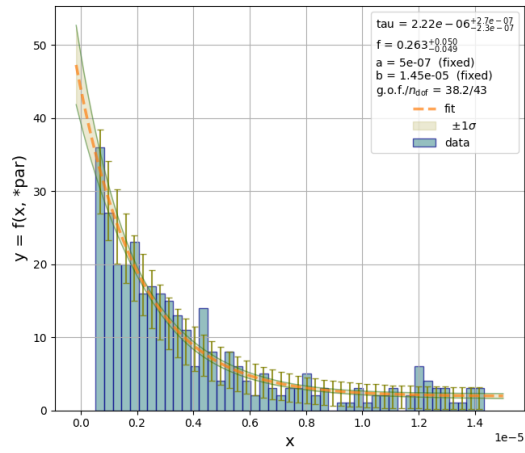


Figure 3.4: Fit of the lifetime to the recorded data

of individual muons were histogrammed. This histogram was fitted with "PhyPraKit". The fit is shown in Figure 3.4.

The lifetime of a muon was found to be approximately $2.22^{+0.27}_{-0.23}\mu s$ which is close enough to the literature value of $2.2\mu s$.

The negative LogLikelihood is shown in Figure 3.6. One observes that the errors are asymmetric. If more events were recorded the errors could be improved and would become more symmetric.

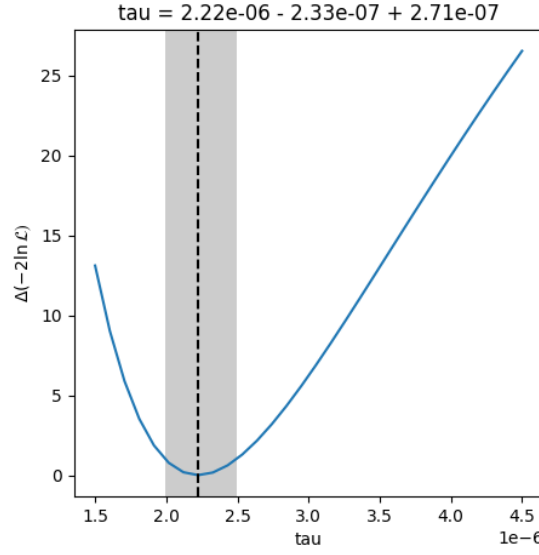


Figure 3.5: Negative LogLikelihood of lifetime measurement fit

3.3 Procedure for Detecting Muon Spin

The following section is based on the provided Jupyter notebook for the muon lifetime experiment [Jup]. As an alternative to the simple exponential decay law, a parameterization of the number of decays detected in the upper detector at time t , which includes a small additional oscillating component (see experiment instructions in the Blue Book) is used:

$$\text{pdf}_{\text{signal}}(t) = K \cdot e^{-t/\tau} \cdot [1 + \bar{A} \cdot \cos(\omega t + \delta)] \quad (3.1)$$

To use this probability density in a fit, it must be normalized over the sensitive time interval $t \in [a, b]$. The required normalization constant K is computed via:

$$K^{-1} = \tau \left(\frac{\bar{A} [e^{-a/\tau} (\cos(a\omega + \delta) - \tau\omega \sin(a\omega + \delta)) + e^{-b/\tau} (\tau\omega \sin(b\omega + \delta) - \cos(b\omega + \delta))]}{\tau^2\omega^2 + 1} + e^{-a/\tau} - e^{-b/\tau} \right) \quad (3.2)$$

Background Component

The background is modeled as a constant probability density in the same time interval:

$$\text{pdf}_{\text{bkg}}(t) = \frac{1}{b - a} \quad (3.3)$$

With a background fraction f , the total probability density function becomes:

$$\text{pdf}(t; \tau, f, \bar{A}, \omega, \delta, a, b) = (1 - f) \cdot \text{pdf}_{\text{signal}}(t) + f \cdot \text{pdf}_{\text{bkg}}(t) \quad (3.4)$$

Phase Shift for Lower Detector

For decays detected in the lower detector, located below the copper absorber, the modulation is phase-shifted by 180° . This effect can be modeled by appropriately adjusting the phase parameter δ . With this adjustment, the same functional form can be reused for both detectors.

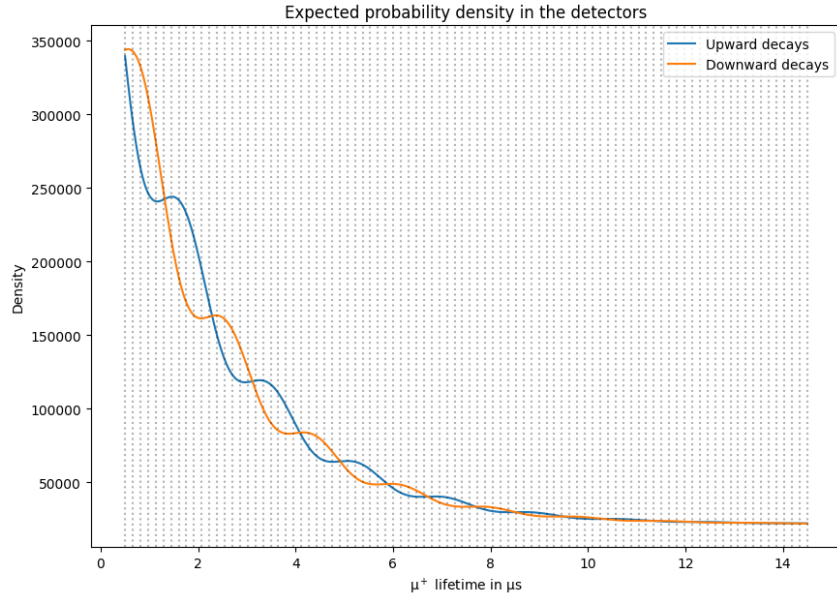


Figure 3.6: Expected probability density

Read in of measurement data and quality control

Unlike the usual one-week measurement period, the data in this case were recorded over a period of two weeks. This allowed for higher than usual trigger heights which should lead to a decreased background.

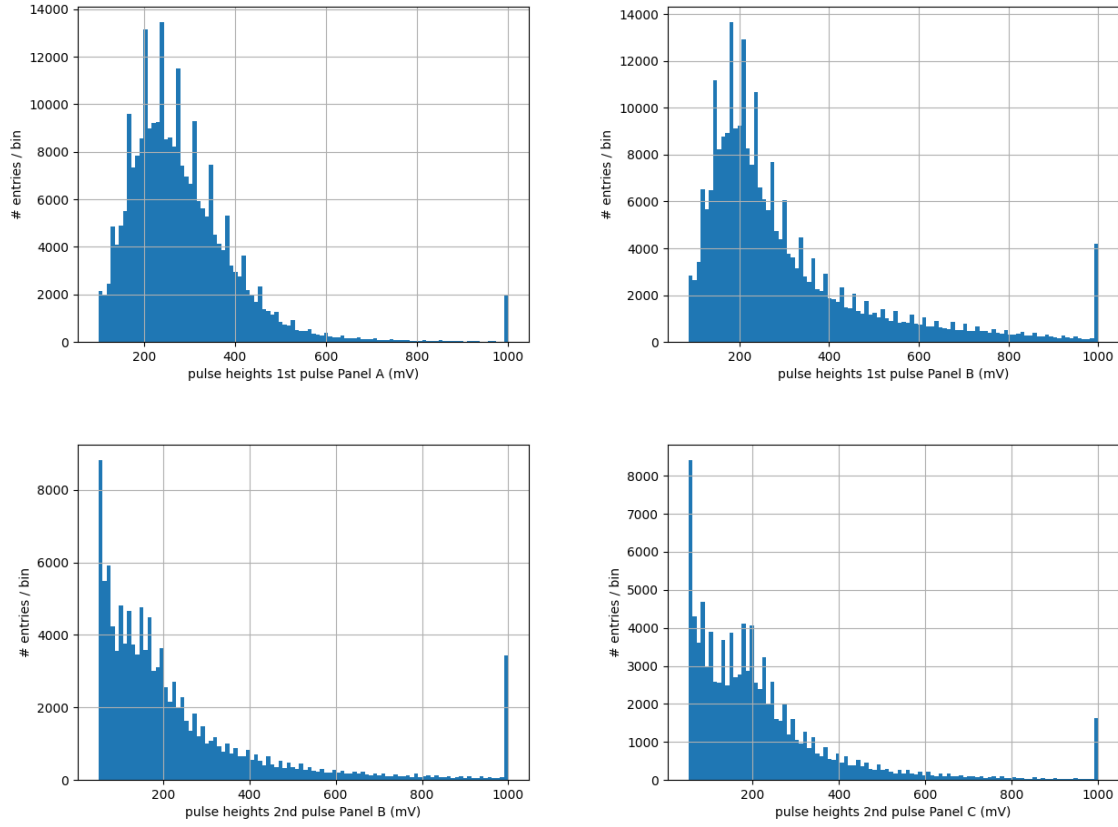


Figure 3.7: histograms of the pulse amplitudes for the primary pulses in panels A and B (top) and the secondary pulses in panels B and C (bottom)

The total number of triggered events is 132 615 274, of which 260 757 pulses were selected for further analysis.

Selection of the measured data

In the analysis of the data, two distinct cases must be considered. When a muon penetrates the copper plate, it decays by emitting a positron, which can be detected either in detector B or detector C. This effectively provides measurements for particles moving upward as well as downward. Consequently, two separate lifetime distributions arise, shown in Figure 3.8. The left panel displays the lifetime distribution for positrons detected by the upper detector (B), while the right panel corresponds to detections beneath the copper plate (detector C). After applying the event selection criteria, 88,766 events for the top detector and 62,746 events for the bottom detector were recorded.

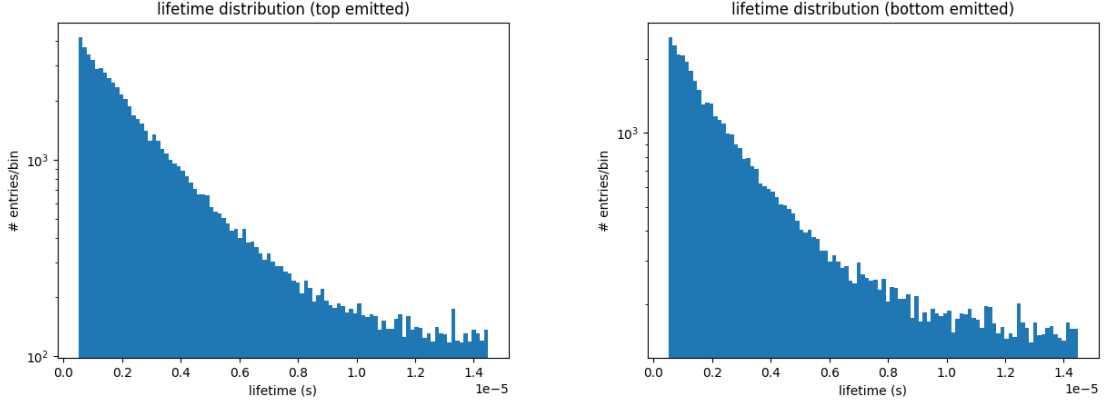


Figure 3.8: lifetime distributions for detector B and C

Adjustments of the pure exponential model ("Null Hypothesis") and the model with spin modulation

Now the distribution of the muon decay times are analysed by fitting two different models to the measured lifetime data. First a pure exponential decay model is applied, representing the null hypothesis with no spin precession. This yields initial estimates for the muon lifetime τ and the background fraction f_{bg} . These fits are on the left hand side of Figure 3.9 and Figure 3.10. In a second step, a refined model including spin modulation is fitted to the data, with τ and f_{bg} fixed values from the first step. This allows the extraction of the asymmetry parameter $\bar{A}$. These fits are on the right hand side of Figure 3.9 and Figure 3.10.

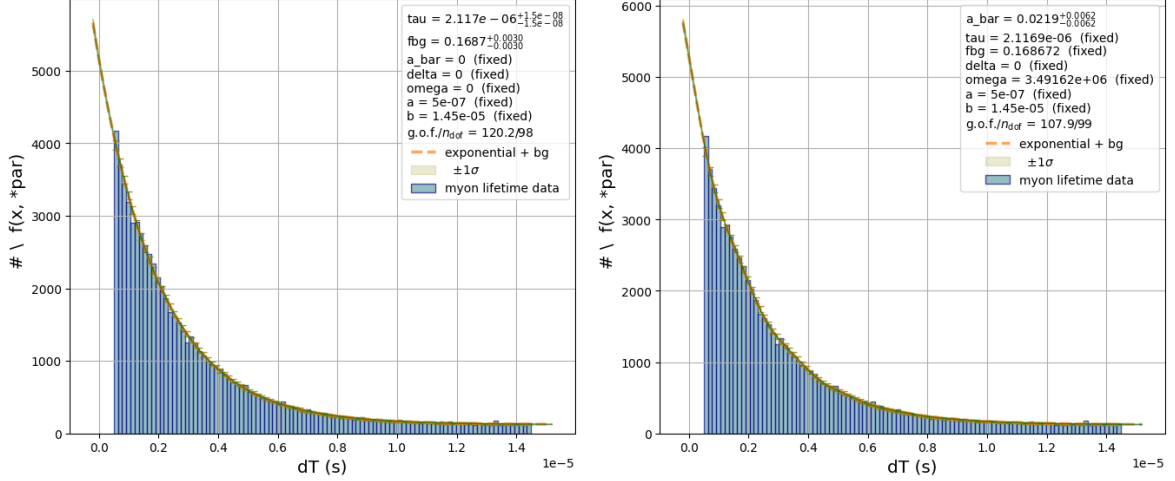


Figure 3.9: Fit of the lifetime to the recorded data of the detector B

For the fit to the data from detector B, the following parameter values were obtained:

- Muon lifetime: $\tau = 2.117 \mu\text{s}_{-0.015}^{+0.015}$
- Background fraction: $f_{bg} = 0.1687_{-0.0030}^{+0.0030}$
- Asymmetry parameter: $\bar{A} = 0.0219_{-0.0062}^{+0.0062}$

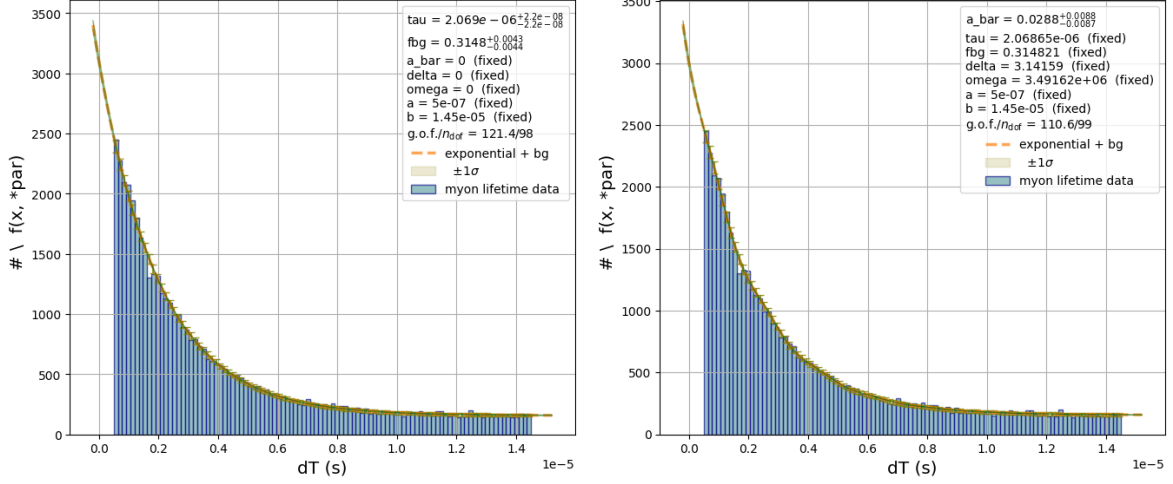


Figure 3.10: Fit of the lifetime to the recorded data of detector C

For the fit to the data from detector C, the following parameter values were obtained:

- Muon lifetime: $\tau = 2.069 \mu\text{s}^{+0.022}_{-0.022}$
- Background fraction: $f_{\text{bg}} = 0.3148^{+0.0043}_{-0.0044}$
- Asymmetry parameter: $\bar{A} = 0.0288^{+0.0088}_{-0.0087}$

Statistical evaluation and determination of the signal significance

Finally the statistical significance of the two different fits, one a pure exponential (null hypothesis), the other one additionally considering the muon precession (alternative precession), is being determined.

The comparison is based on the values of the negative log-likelihood (NLL) from both fits. The test statistic is the difference of these values, defined as

$$z^2 = -2 \cdot \Delta \ln \mathcal{L} = -2 \cdot (\text{NLL}_{\text{mod}} - \text{NLL}_{\text{non-mod}}), \quad (3.5)$$

where a higher z^2 value indicates a stronger preference for the modulation model. The square root z corresponds to the significance of the observed modulation in terms of standard deviations. With z , the corresponding p -value is calculated using the cumulative distribution function Φ of the standard normal distribution:

$$p = 1 - 2(1 - \Phi(z)), \quad z \geq 0. \quad (3.6)$$

The p -value is a measure for the probability that the observed deviation from the null hypothesis is purely due to statistical fluctuations. Small p -values therefore indicate statistically significant signals.

This evaluation helps decide whether the oscillation seen in the data is meaningful.

For both the upper and lower detectors, the fits including modulation yield significantly better agreement with the data compared to the pure exponential model. The statistical results are:

- **Upper Detector (B):** $z = 3.52 \Rightarrow p = 0.02195\%$
- **Lower Detector (C):** $z = 3.29 \Rightarrow p = 0.04925\%$

In both cases, the p -values are well below the common threshold of 5%, allowing us to reject the null hypothesis. This provides strong evidence for the presence of a modulation in the muon decay time distribution, consistent with spin precession.

Quellen

[Jup] Template for the evaluation of the muon experiment

[Wol] WOLF, Joachim: *Introduction to the Nuclear and Particle Physics Lab Course*. – preparation sheet from Ilias

List of figures

Figure 2.1: [Wol], p.206, modified